

Rossmann Store Sales Forecasting

Report 2 — End-to-End Analysis, Deep Learning & Deployment

From customer purchasing behaviour to a production-ready forecasting system

1.017M	1,115	942	0.104
training rows	stores	aggregate daily observations	LSTM RMSPE

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Project: Sales Forecasting Across Multiple Retail Stores — Rossmann Pharmaceuticals

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1. Executive Summary

Rossmann sales behaviour is driven by a combination of customer traffic, promotions, calendar effects, store characteristics and local competition. The analysis therefore progresses from exploratory analysis to a baseline Random Forest, an improved LightGBM model, and a two-layer LSTM sequence model.

Headline result: the LSTM achieved 0.1041 RMSPE and 577.3 MAE on the held-out six-week aggregate validation window. The production dashboard uses LightGBM instead because its prediction granularity is per store and per day, matching the operational requirement for store managers.

0.343	0.1313	0.1041	0.111
v1 Random Forest RMSPE	v2 LightGBM RMSPE	LSTM RMSPE	dashboard Customers RMSPE

Business decision

- Further investigation is worthwhile: promotion response varies substantially by store and should be investigated at store level.
- Customer traffic is a major commercial signal: Sales–Customers correlation is 0.824.
- Weekly seasonality is dominant in the aggregate series, making calendar-aware and sequence-based models valuable.
- LightGBM is the practical production choice; LSTM is a strong complementary aggregate-demand model.

The final assignment asks for a blog-style PDF, GitHub notebook link, deployed application evidence and screenshots demonstrating additional work.

2. Data, Cleaning & Feature Engineering

The EDA workflow loaded 1,017,209 training rows, 41,088 test rows and 1,115 store records. Store metadata was merged into the transaction tables, producing 18 training columns.

Missing-value handling produced zero remaining nulls in both train and test in the recorded EDA run. Closed-store observations were retained separately but excluded from open-store sales-behaviour averages. Potential extreme sales values were flagged using a $3\times\text{IQR}$ rule rather than automatically deleted.

The forecasting feature pipeline extracts Year, Month, Day, WeekOfYear, IsWeekend and MonthPeriod. It also derives DaysToHoliday, DaysAfterHoliday, CompetitionOpenMonths, Promo2OpenWeeks and IsPromo2Month.

This design keeps the model focused on signals that are available from the date and store metadata while avoiding future leakage in the target-encoding step.

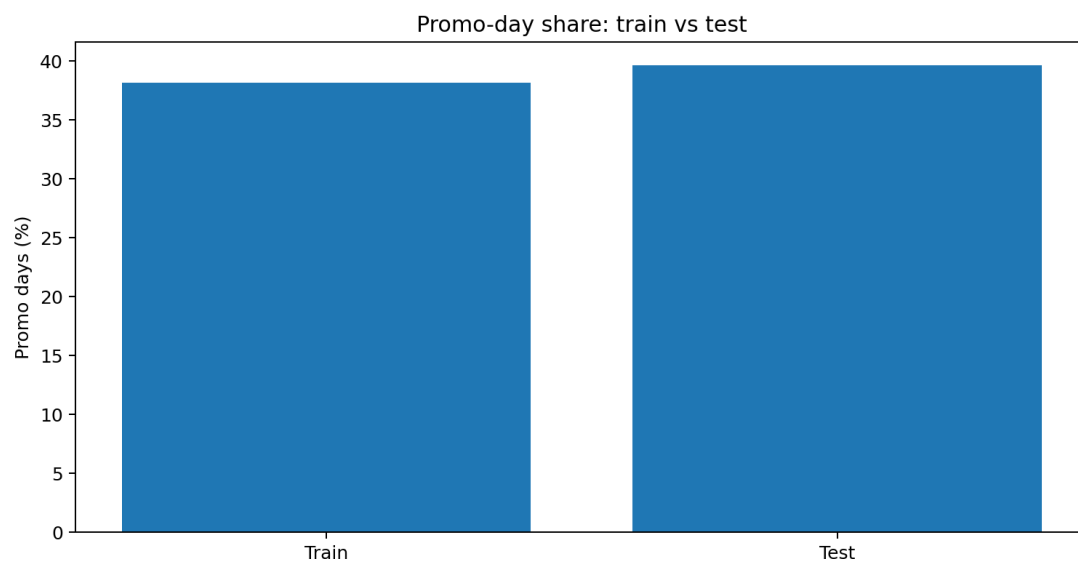


Figure 1. Promotion exposure is closely aligned between train (38.15%) and test (39.58%), reducing concern about promo-distribution shift.

3. Customer Purchasing Behaviour & Promotion Effects

The EDA calculates a Pearson correlation of **0.824** between Sales and Customers. This supports a simple commercial interpretation: stores with more customer traffic generally generate more sales.

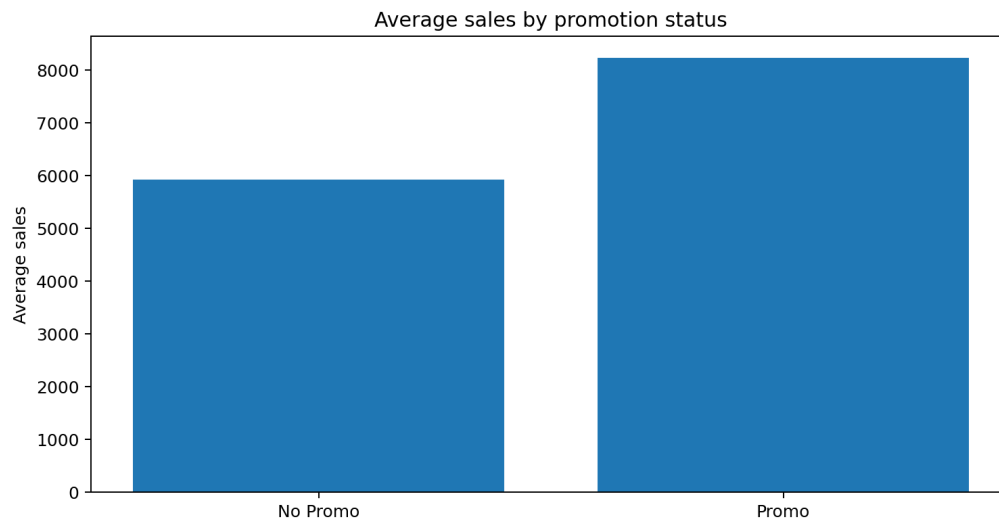


Figure 2. Average sales increase from 5,929.41 without a promo to 8,228.28 on promo days.

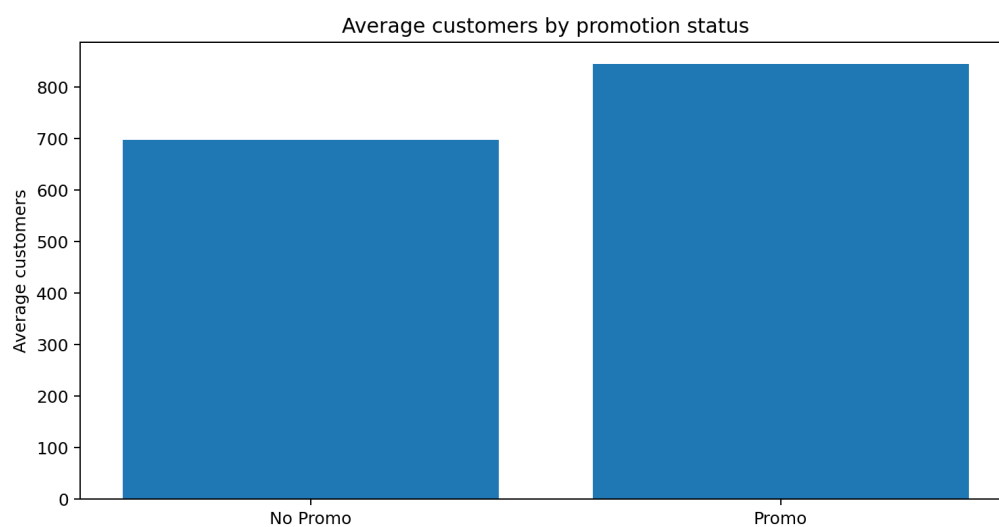


Figure 3. Average customers increase from 696.86 to 844.43 on promo days. Sales per customer also rises from 8.51 to 9.74.

The strongest observed store-level promotion uplift is 144.93%, with several other stores showing uplift above 90%. This is a strong signal for targeted promotion analysis rather than a one-size-fits-all policy.

4. Holidays, Seasonality & Store Context

Holiday behaviour shows a clear pre-holiday purchasing effect. The EDA describes Christmas and Easter as the strongest seasonal peaks, with sales tending to rise before holidays and soften after them. State-holiday labels also show higher average sales for Easter and Christmas than ordinary non-holiday days.

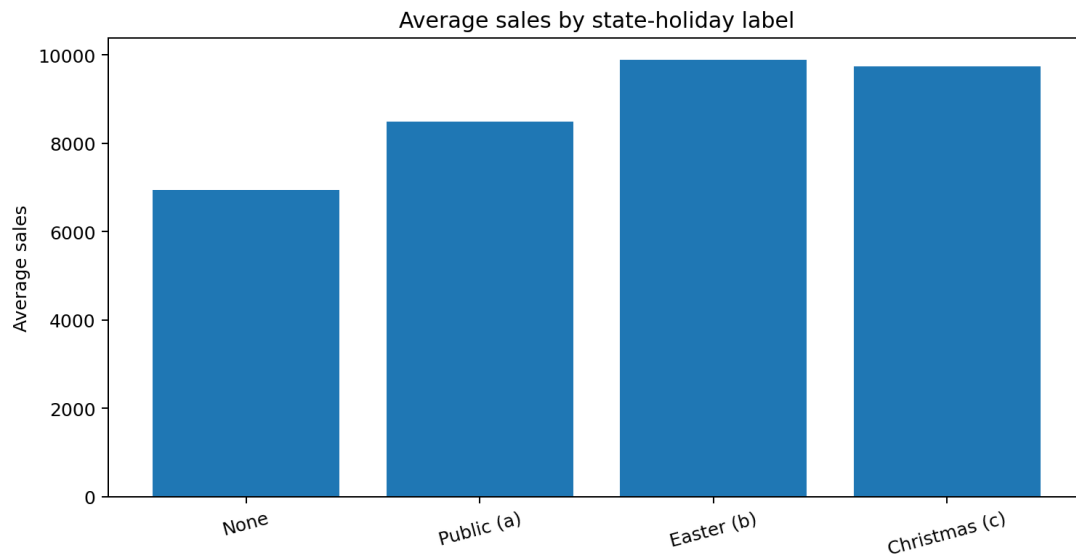


Figure 4. Average sales by state-holiday label. The observed open-store averages are shown; many stores are closed on state holidays, so this is not a causal holiday-lift estimate.

Store assortment is associated with different sales levels: basic 6,621.02, extra 8,639.35 and extended 7,300.53 average sales. The extra group is small, so this is directional rather than causal.

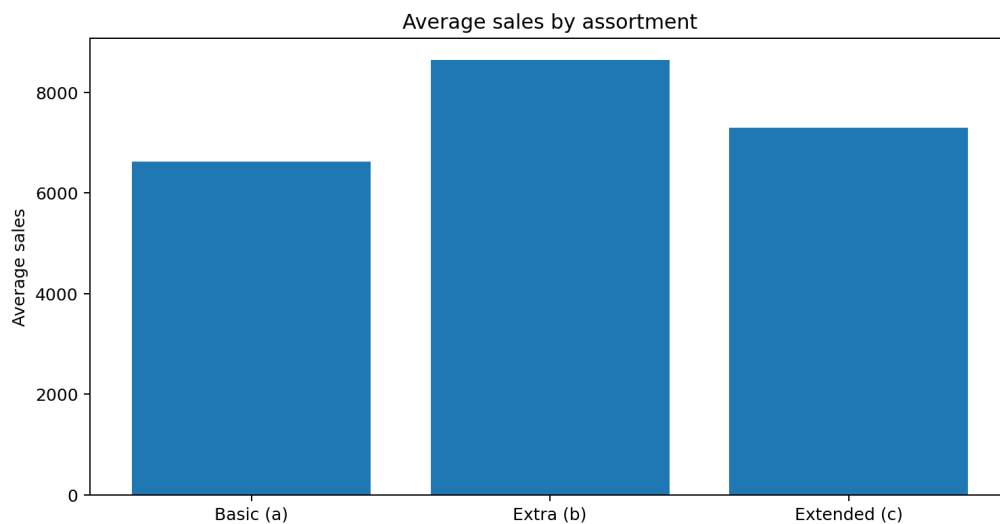


Figure 5. Average sales by assortment type.

5. Competition: A Contextual, Not Monotonic, Signal

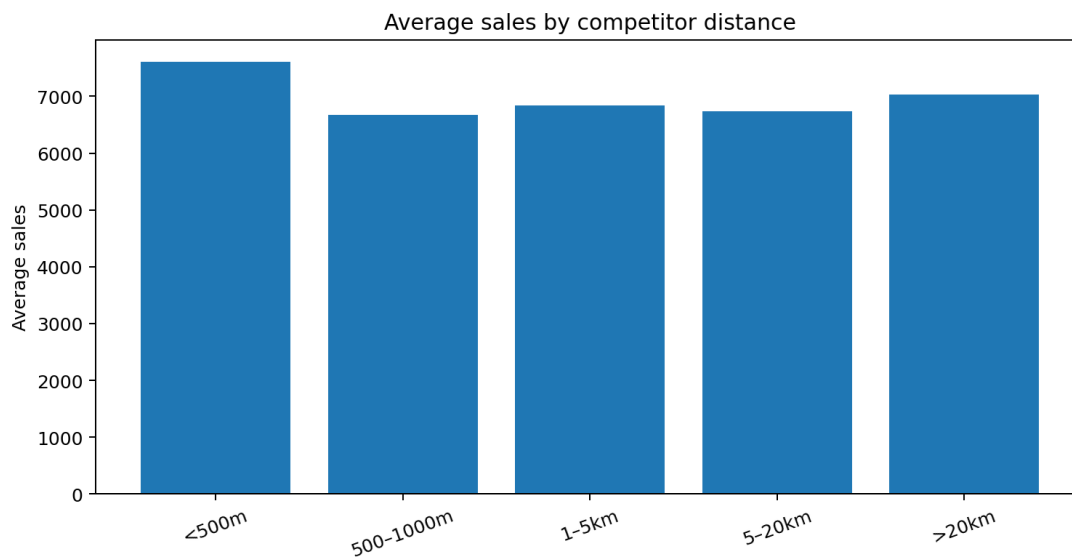


Figure 6. Competitor distance does not have a simple monotonic relationship with sales. The <500m bucket averages 7,609.90 sales, while the 500–1000m bucket averages 6,676.22.

This counter-intuitive result is commercially plausible: a store close to a competitor may also be located in a high-footfall commercial area. Therefore CompetitionDistance should be interpreted together with StoreType, Assortment and other locality proxies.

Competitor openings are similarly heterogeneous. For 188 stores with usable impact estimates, the median percentage sales change after a competitor opening is -0.17% , while the observed range spans -32.11% to $+25.09\%$. That wide spread argues against a universal competitor-opening rule.

Model implication

Competition variables should be included as contextual features rather than used as standalone causal explanations.

6. Machine Learning Baseline → LightGBM v2

The first machine-learning implementation used a scikit-learn Pipeline with preprocessing and Random Forest regression. The validation protocol held out the last six weeks of the training period, matching the intended six-week forecast horizon.

RMSPE was selected as the primary metric because it is the task's percentage-error measure and is more scale-aware across stores of very different sales volumes. MAE and R^2 were reported alongside it.

The v2 LightGBM model introduced leak-safe, smoothed StoreTargetEnc. It replaces the raw store integer ID with historical average sales computed from the training split only. This feature became the most important signal by gain.

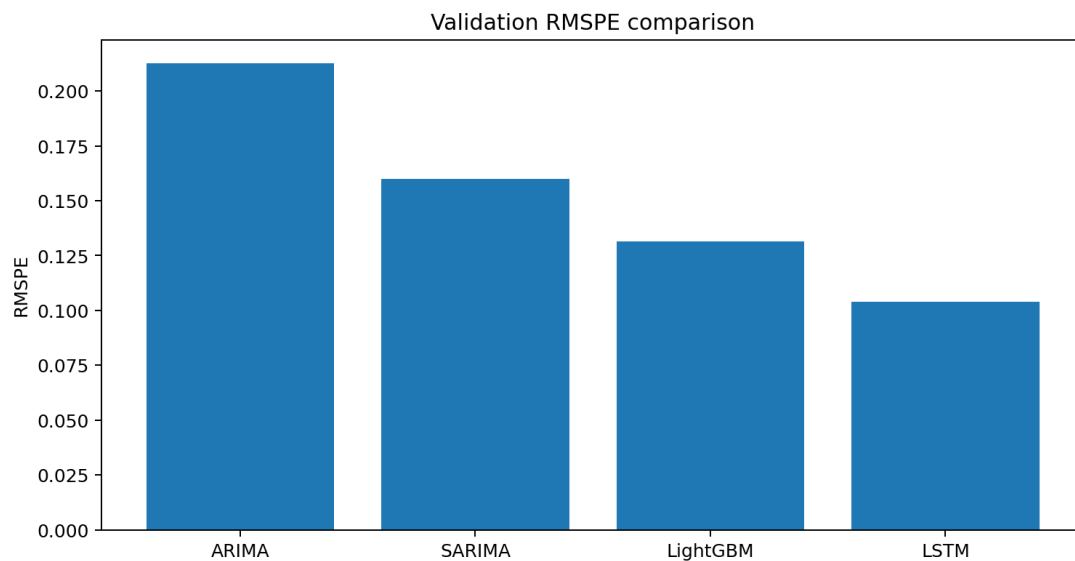


Figure 7. Validation RMSPE across the compared forecasting approaches.

v1 Random Forest: RMSPE 0.343, MAE 1,523, R^2 0.50. v2 LightGBM: RMSPE 0.131, MAE 614, R^2 0.915 — a 62% RMSPE reduction.

7. Deep Learning Focus — LSTM Design

The deep-learning notebook isolates the aggregate daily mean sales per open store, producing 942 observations from 1 January 2013 through 31 July 2015. The purpose is to create a clean sequence for comparing classical time-series models and the required deep-learning model.

An Augmented Dickey–Fuller test gives a statistic of -5.4154 with $p=0.0000$, so the raw series is treated as stationary and no differencing is applied. ACF/PACF reveal a strong lag-7 pattern, confirming weekly seasonality.

The LSTM converts the series into supervised sliding-window data using the previous 14 days to predict the next day. Scaling is fitted on training data only, preventing leakage.

Component	Configuration
Architecture	LSTM(32, tanh, return_sequences=True) → LSTM(32, tanh) → Dense(1)
Input window	14 days
Parameters	12,705 trainable parameters
Optimizer / loss	Adam / MSE
Training	Up to 80 epochs, batch size 16
Stopping	Early stopping, patience 8, restore best weights
Recorded training	Stopped after 79 epochs

The recorded architecture is explicitly a two-layer LSTM with 32 units in each recurrent layer and a Dense(1) output, totaling 12,705 trainable parameters. Early stopping stopped the recorded run after 79 epochs.

8. Deep Learning Results & Model Comparison

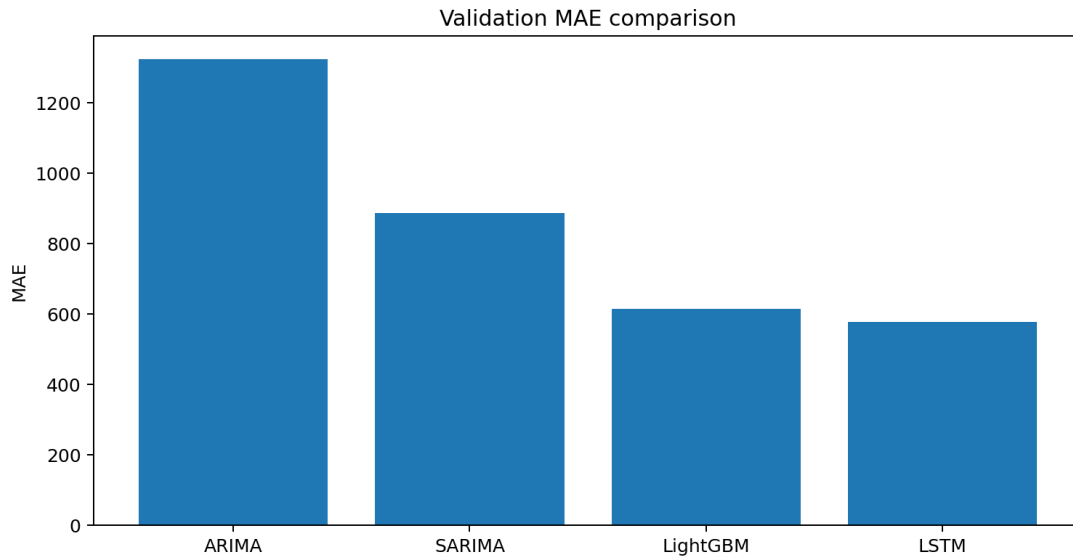


Figure 8. Validation MAE comparison on the same six-week aggregate window.

Model	Granularity	RMSPE	MAE
ARIMA(2,0,2)	Aggregate daily	0.2125	1322.0
SARIMA(2,0,2)x(1,1,1,7)	Aggregate daily	0.1600	885.8
LSTM (2-layer, 14-day window)	Aggregate daily	0.1041	577.3
LightGBM + Store target encoding	Per-store daily	0.1313	614.0

The LSTM is the best numerical result in the aggregate experiment: RMSPE 0.104083 and MAE 577.26. It beats SARIMA and ARIMA on the same six-week aggregate validation window.

Critical comparability caveat: the LSTM predicts one aggregate number per day across all stores, while LightGBM predicts store-day sales. Therefore the lower LSTM RMSPE does not make it the better production model for a manager asking what a specific store will sell next Tuesday.

Operational interpretation

- LSTM → useful for aggregate demand, capacity and inventory planning.
- LightGBM → useful for individual-store forecasts and operational decisions.

9. Explainability & Prediction Intervals

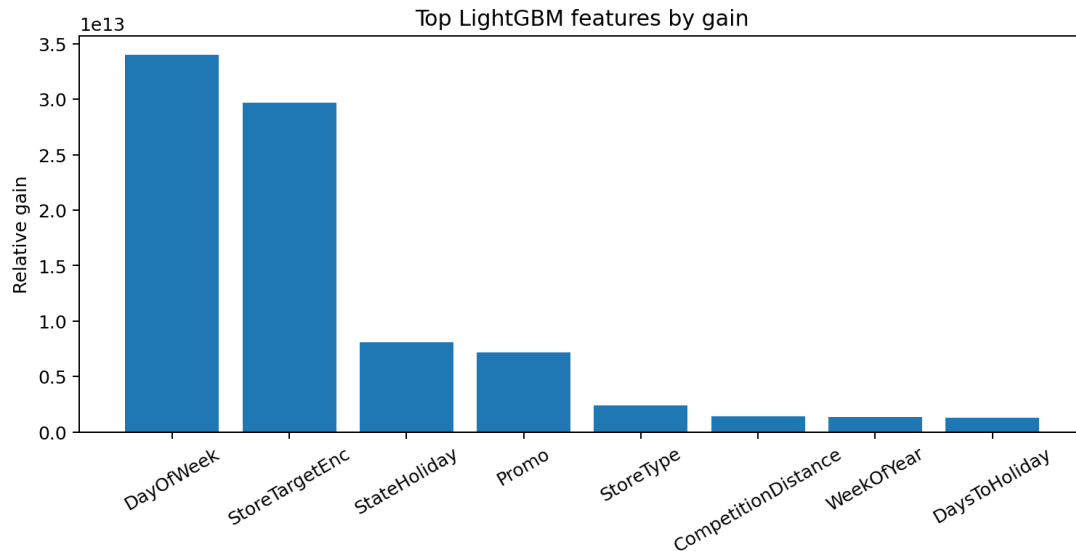


Figure 9. LightGBM feature gain. DayOfWeek and StoreTargetEnc are the strongest reported signals, followed by StateHoliday, Promo, StoreType and CompetitionDistance.

The feature-importance result reinforces the EDA story: weekly timing and store-level historical demand are central to sales forecasting. Promotions and holidays are important operational variables, while competition contributes additional context.

For uncertainty, the production LightGBM bundle trains two quantile models at the 5th and 95th percentiles, producing a 90% prediction band around the point forecast.

Example validation output: for an actual sales value of 13,995, the point prediction was 12,222, with a 5th–95th percentile interval of approximately 9,413–14,946. This communicates a range rather than only a single estimate.

10. Deployment — Streamlit Forecasting Dashboard

The final dashboard is deployed on Streamlit and provides two workflows: single-store/date-range forecasting and CSV batch prediction. It displays model validation metrics, accepts business inputs, plots predicted sales and customers, and supports CSV download.

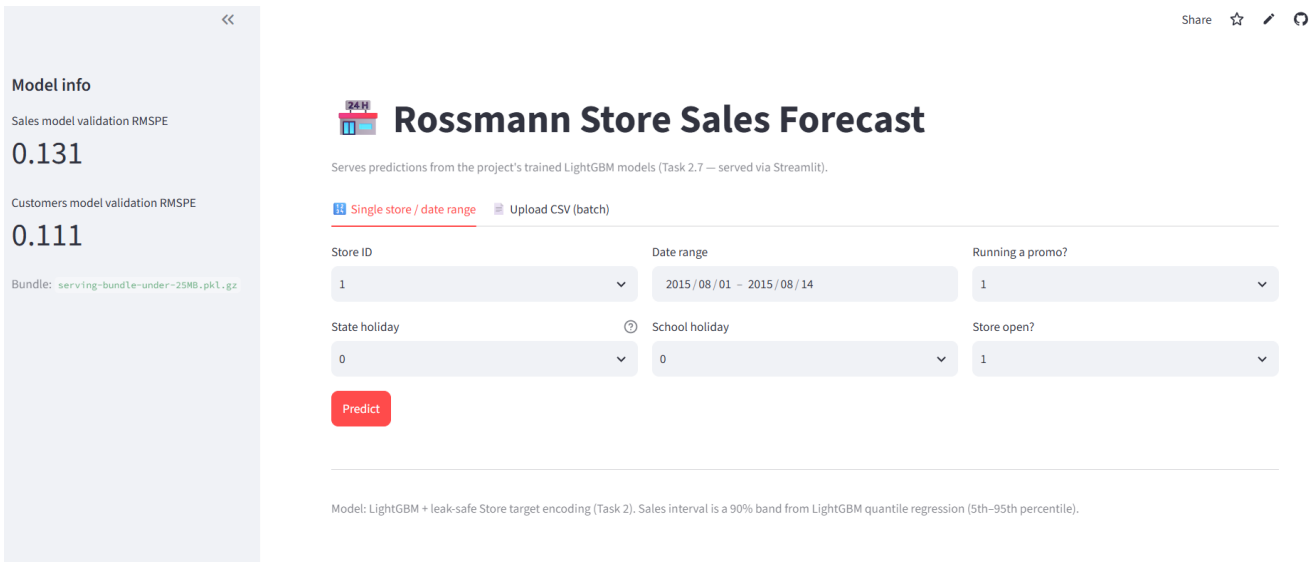


Figure 10. Deployed Rossmann Store Sales Forecast dashboard. The screenshot shows Sales RMSPE 0.131, Customers RMSPE 0.111 and the active compressed serving bundle.

The serving application loads the serialized bundle, merges store metadata, applies the same feature engineering and store target encoding used during training, generates sales and customer predictions, and forces predictions to zero when a store is closed.

GitHub: https://github.com/shakya362/rossmann_store_sales

Deployed application: <https://rossmann-store-sales-forecast.streamlit.app/>

11. Model Serialization, Versioning & Evidence

The project serializes models with timestamps so successive trained artifacts can be retained and compared. The LightGBM bundle includes the point estimator, 5th/95th percentile quantile models, store encoding map, global mean and feature definitions. The LSTM is serialized separately as a timestamped .keras model with metadata including its window, training range, validation RMSPE/MAE and number of epochs.

This creates a reproducible path from experiment → serialized artifact → deployed service.

Evidence	Status	Submission action
GitHub notebooks	Available	Use repository link
Deployed dashboard	Available	Use URL + screenshot
Timestamped LightGBM bundle	Available	Retain artifact evidence
Timestamped LSTM artifact	Available	Retain .keras + metadata
DVC multiple data versions	Not evidenced by supplied files	Add actual DVC screenshot
MLflow multiple model versions	Not evidenced by supplied files	Add actual MLflow screenshot if required

No DVC or MLflow screenshot was fabricated because none was supplied in the project materials. Add the real screenshots before final submission if those items are mandatory.

12. Conclusion & Recommended Next Steps

The project demonstrates an end-to-end forecasting workflow in which EDA informs feature engineering, model selection and deployment. The first-pass analysis shows clear weekly, holiday and promotion effects, while customer traffic and store-level historical demand emerge as especially strong predictors.

The deep-learning experiment is the report's strongest technical highlight. A compact two-layer LSTM learns the weekly structure from a 14-day sequence and reaches 0.1041 RMSPE on the aggregate six-week validation window. This outperforms the aggregate ARIMA and SARIMA benchmarks.

For production, however, the per-store LightGBM remains the recommended model because it directly answers the store manager's operational question and uses richer contextual features. The LSTM and LightGBM should therefore be viewed as complementary rather than competing models.

Recommended next experiments

- Add previous-week same-weekday lag and rolling-window sales features at store level.
- Develop a dedicated Customers forecasting model and use predicted traffic as a sales signal.
- Measure store-specific promotional uplift and use it to prioritize promotion allocation.
- Explore a hybrid/ensemble design combining store-level LightGBM with aggregate LSTM forecasts.
- Track new timestamped model versions on the same time-based validation protocol.

Final submission checklist

- PDF report
- GitHub repository containing Jupyter notebooks
- Deployed Streamlit application link
- Dashboard screenshot
- DVC multiple-data-version screenshot
- MLflow multiple-model-version screenshot, if required

Appendix — Technical Reference

EDA: 1,017,209 training rows; 41,088 test rows; 1,115 stores; zero remaining nulls after the recorded cleaning workflow.

Promotion: train 38.15%; test 39.58%. Promo sales 8,228.28 vs 5,929.41 without promo; promo customers 844.43 vs 696.86.

Customer relationship: Sales–Customers correlation 0.824.

Time series: 942 aggregate daily observations; ADF statistic -5.4154 , $p=0.0000$; 42-day validation window.

LSTM: 14-day window; two LSTM layers of 32 units; Dense(1); Adam/MSE; batch size 16; up to 80 epochs; early stopping patience 8; 12,705 trainable parameters.

Results: ARIMA 0.2125 RMSPE / 1322.0 MAE; SARIMA 0.1600 / 885.8; LSTM 0.1041 / 577.3; LightGBM 0.1313 / 614.0.

Production uncertainty: LightGBM 5th/95th quantile models provide a 90% prediction band.

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